

Lecture Notes: Evaluation & Optimization in LLM & Prompt Engineering

1. Prompt Evaluation Metrics

When designing prompts, you must judge whether the model's response is *good*. The three key metrics are **Relevance**, **Coherence**, **Accuracy**.

(a) Relevance

- **Definition:** Does the output match the intent of the prompt?
- **Analogy:** Asking your friend "*What time is the train?*"
 - ✓ Relevant: "*9:15 AM.*"
 - ✓ Irrelevant: "*I love trains.*"

(b) Coherence

- **Definition:** Is the response logical, clear, and structured?
- **Analogy:** Like reading a well-written essay vs. one that jumps randomly between topics.

(c) Accuracy

- **Definition:** Factual correctness of the answer.
- **Problem:** LLMs can "hallucinate" facts.
- **Analogy:** A confident student who gives wrong answers.

(d) Exercise 1

- ✧ Prompt: "*Summarize NEP 2020 in 3 points.*"
 - ✓ Output A: "*NEP 2020 focuses on holistic learning, vocational training, and flexible curriculum.*"
 - ✓ Output B: "*India's population is very young. Education is important for the economy.*"
 - ✧ Q1: Which output is more **relevant**?
 - ✧ Q2: Which output is more **coherent**?
 - ✧ Q3: Which output is more **accurate**?
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2. Latency Measurement & Performance Profiling

(a) Latency = Response Time.

- **Analogy:** Ordering food at a restaurant. A fast restaurant = low latency; a slow one = high latency.
- **Why Important?**

Real-time applications (chatbots, voice assistants, trading bots) need low latency.

(b) Profiling Breakdown

- ✓ **Input Processing Time** – reading your prompt.
- ✓ **Token Generation Time** – producing words/tokens per second.

(c) Optimization

- ✓ Keep prompts short.
- ✓ Use smaller models for simple tasks.
- ✓ Cache frequent responses.

(d) Exercise 2

You ask the same question to two chatbots.

- ✓ Bot A → responds in 1.2 seconds.
- ✓ Bot B → responds in 4.8 seconds.
- ✧ Q1: Which bot has lower latency?
- ✧ Q2: If both are equally accurate, which one would you deploy for a customer service app? Why?

3. Token Usage & Cost Estimation

(a) Tokens = Units of text (not full words).

(b) Analogy: Like SMS charges. A long text = more SMS units = more cost.

(c) Example

- ✓ Prompt: “Summarize NEP 2020 in 3 points.” → 20 tokens.
- ✧ Response: 60 tokens.
- ✧ Total = 80 tokens.
- ✧ If cost = \$0.01 per 1000 tokens → $80/1000 \times \$0.01 = \0.0008 .

(d) Optimization

- ✧ Remove unnecessary words.
- ✧ Store context in embeddings instead of repeating it.

(e) Exercise 3

- ✓ Prompt A = 50 tokens, Output = 150 tokens.
 - ✓ Prompt B = 15 tokens, Output = 80 tokens.
 - ✧ Q1: Calculate total token usage for each case.
 - ✧ Q2: If cost = \$0.02 per 1000 tokens, which case is cheaper?
 - ✧ Q3: How could you optimize Prompt A to reduce cost?
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4. Fallback Strategies & Multi-Model Orchestration

(a) Fallback Strategies

- **Definition:** Backup plan when main model fails.
- **Analogy:** Spare tyre for your car.

(b) Multi-Model Orchestration

- **Definition:** Different models handle different tasks.
- **Analogy:** Hospital team → Nurse (triage), Doctor (diagnosis), Specialist (complex cases).

(c) Example

- ✓ Pipeline:
 - ✧ Small model → classifies if input is a question, translation, or summarization.
 - ✧ Large model → handles only complex queries.

(d) Benefits

Lower cost, faster, more reliable.

(e) Exercise 4

- ✓ Suppose GPT-4 costs \$0.06 per 1K tokens, GPT-3.5 costs \$0.002 per 1K tokens.
 - ✧ Q1: If a chatbot always uses GPT-4 for 10K tokens/day, what is the daily cost?
 - ✧ Q2: If you use GPT-3.5 for 70% of queries and GPT-4 for only 30%, what is the new cost?

✧ Q3: Which strategy is more efficient?

5. Final Project Demo, Feedback & Peer Review

(a) Final Demo

- Students present their LLM application.
- Must show: prompt design, evaluation metrics, latency, token cost, fallback strategy.

(b) Feedback

- Instructor + peers evaluate:
 - ✓ Relevance, coherence, accuracy.
 - ✓ Speed & cost efficiency.
 - ✓ Innovation in design.

(c) Peer Review

- Students evaluate each other's projects with rubrics.
- **Analogy:** Like peer correction of exam papers.

(d) Exercise 5 (Project Activity)

- Design a **mini LLM system** (chatbot, summarizer, or translator).
- Document:
 - ✓ Your **prompt design**.
 - ✓ How you measure **relevance, coherence, and accuracy**.
 - ✓ **Latency profiling** results.
 - ✓ Estimated **token cost**.
 - ✓ Whether you used **fallback or orchestration**